

Erdős Problem #345

EP-345

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A \subseteq \mathbb{N}$ be a complete sequence, and define the threshold of completeness $T(A)$ to be the least integer m such that all $n \geq m$ are in $P(A) = \{\sum_{n \in B} n : B \subseteq A \text{ finite}\}$ (the existence of $T(A)$ is guaranteed by completeness).

Is it true that there are infinitely many k such that $T(n^k) > T(n^{k+1})$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>